Frederik Dufour

Website | Email | LinkedIn | (514)-917-5282

Economist and Data Scientist - Energy markets, Regulation, Financial Economics, Large Datasets

Languages: English (*Fluent*), French (*Native*)

EXPERIENCE

Ph.D. Candidate (Researcher) – Economics, University of Toronto **2023-Present**

Researcher with a varied portfolio of energy-related projects leveraging modern quantitative and programming techniques.

- **Review of Total Cost Benchmarking Methodology for Electricity Distributors** *with A. Yatchew et. al. (2024)*
  Published on the OEB website
  Compared international regulatory practices and benchmarking methods to inform the OEB's assessment of electricity distributor efficiency.
- **Electricity Distribution in the Energy Transition: Revisiting Productivity Growth in Ontario**
  joint with A. Yatchew and M. Scafe
  Assessed Ontario distributors' productivity growth using cost models that account for distributed generation and electrification, while providing causal evidence that mergers' effects on productivity were limited.
- **Electricity Futures and Time-Varying Risk Premia: Evidence from Alberta** *joint with D. Olmstead and A. Yatchew*
  Developed a semi-structural model of Alberta electricity price expectations to estimate information propagation.
- **Investment Timing Under Regulatory Cycles**
  Examined whether regulatory rate resets induce Ontario electricity distributors to concentrate capital investment.
- **Power Outages and Household Consumption - Evidence from California**
  Leveraged satellite nighttime lights to estimate how California's preventive power shutoffs affected consumption.

Research Assistant **2019-2024**

- Collaborated on 5+ projects
- Produced code to handle large datasets (50+ GBs) used to replicate various papers (Kroft et al. (2022) and DellaVigna & Gentzkow (2020)).
- Data analysis for topics related to "Social Transfers in Kind" (STiK) to provide stronger measurement of consumption inequality, COVID-19 federal and provincial measures, costs of pipe leakage in cities.

Teaching Assistant **2021-2026**

- Produced various data visualizations using geospatial data.
- Energy and Regulation, Energy and the Environment, Environmental Economics and Policies, Software Head TA (Python,R, Stata, MATLAB).

EDUCATION

PhD – Economics, University of Toronto **2027 (Expected)**

- Committee: Adonis Yatchew (Supervisor), Derek Olmstead, Jacquelyn Pless
- Awards: OEB-sponsored Fellowship ($15K), Faculty of Arts and Science Doctoral Fellowship ($70K), Ph.D. Direct Access Fellowship ($10K).

M. Sc. – Economics, Université de Montréal **2021**

- Awards: Lise Salvas Foundation Award ($1K), Maurice Bouchard Foundation Award ($1K)

Honours B. Sc. – Economics, Université de Montréal (cGPA: 4.0) **2020**

- Awards: Jean-Marie-Dufour Award ($500), Roger Dehem Award ($500), Robert-Lacroix Award ($500)

SKILLS

Programming: Python (advanced), R (advanced), Stata (advanced), MATLAB (proficient), LaTeX (advanced)
Tools: QGIS, Data Processing, Data Visualization, Statistics, Econometrics, Machine Learning, Big Data Optimization, Semi-parametric Modelling, Version control (Git, Github), AI (OpenAI, Anthropic).
Professional: Collaboration, Detail-oriented, Report writing.

INVOLVEMENT

Student Energy (UofT Chapter), production of weekly energy-related info capsules **2024-2025**

Participant, The Governor's Challenge (Bank of Canada) **2020**